Suspended Sediment Particle Size Prediction

Random Forest, Support Vector Regression, and XGBoost

Problem statement

This project estimates two properties of suspended-sediment particle-size distributions from environmental measurements: median particle diameter (d50) and distribution variance (sigma2). It compares three regression models using the supplied notebook workflow, with the evaluation corrected so held-out target labels do not influence filtering or feature selection.

Dataset and preparation

The supplied *synthetic_data_improved.csv* contains 1,648 hourly rows, eight environmental predictors, and the two target columns. Predictor columns contain missing entries. This is the provided synthetic dataset; it is not verified field data.

The time column was excluded and incomplete rows were dropped, leaving 1,402 samples. The data were shuffled into 70% training and 30% test sets with random_state=42 (981 and 421 rows). No target-based outlier filtering was applied. Random Forest feature importance was fitted on training rows only; the four selected predictors were ub, S, chl_a, and T.

Approach and implementation

Separate regressors were fitted for d50 and sigma2. Random Forest used 400 trees. SVR used standardized inputs and a five-fold grid search over C and epsilon, performed only on training data. XGBoost used 300 trees, depth 3, and learning rate 0.05. Each model was evaluated once on the held-out test set using R^2, root mean squared error (RMSE), and mean absolute error (MAE). The executed notebook reports the same workflow, comparison table, and plots.

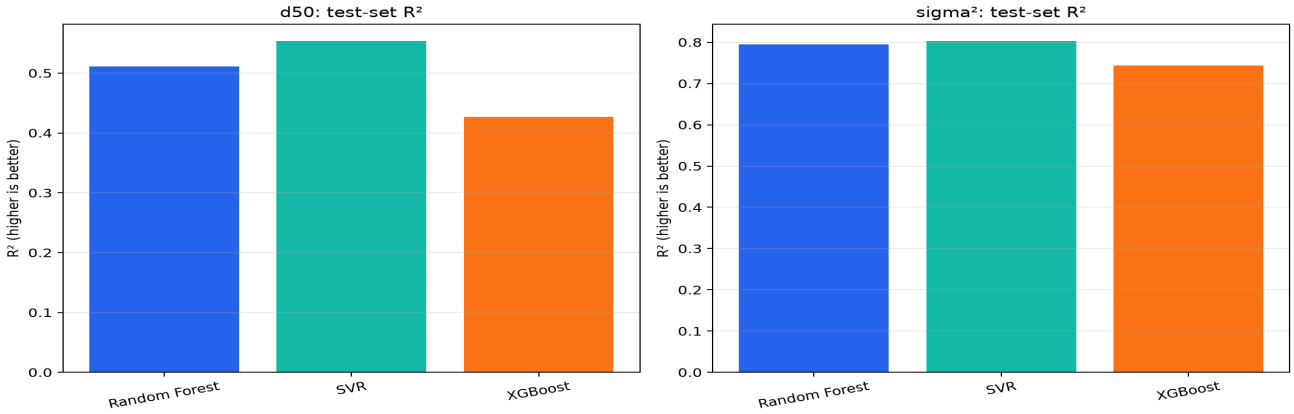
Held-out test results

Target	Model	R^2	RMSE	MAE
d50	Random Forest	0.511284	38.4244	15.8331
sigma2	Random Forest	0.795068	13.4632	6.9932
d50	SVR	0.553691	36.7195	13.9130
sigma2	SVR	0.803180	13.1941	6.2536
d50	XGBoost	0.427198	41.5988	16.4480
sigma2	XGBoost	0.744005	15.0474	7.6167

Higher R^2 and lower RMSE/MAE indicate better fit on this particular test split.

Findings and project notes

Results from the corrected notebook workflow



Conclusion

SVR achieved the highest test R² in this run for both targets: 0.553691 for d50 and 0.803180 for sigma2. Random Forest scored 0.511284 and 0.795068, while XGBoost scored 0.427198 and 0.744005, respectively. These results describe this synthetic dataset and the specified random holdout; they do not establish performance on measured sediment samples.

Limitations

The data are synthetic and may not represent field behavior. A random row split evaluates performance on similar generated samples and is not a future-period forecast. Extreme target values were retained rather than removed using held-out labels. Results can vary with software versions and should be regenerated by running the project.

Reproducibility

The project folder contains the executed step-by-step notebook, the supplied dataset, requirements.txt, README setup instructions, run_project.py, saved metrics and predictions, and this report. From the project root, install dependencies with `pip install -r requirements.txt`, then run `python run_project.py`. The notebook and script use the same split, training-only feature selection, model fitting, and evaluation.

Team members: _____ _____